LESSON 116 (1.0 GROUND and 1.5 DUAL)

Lesson Objective

During this lesson, the student will gain proficiency with ground reference maneuvers, emergency procedures and normal traffic pattern procedures. The student will practice steep turns while descending to understand how to perform an avoidance turn while maintaining a descent.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
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- Student questions.

References

- Aeronautical Information Manual. (AIM)
    - Chapter 03 - Airspace.
        - Section 02 - Radio Communications Phraseology and Techniques.
        - Section 03 - Airport Operations.
    - Chapter 06 - Emergency Procedures.
- Airplane Flying Handbook. (FAA-H-8083-3C)
    - Chapter 07 - Ground Reference Maneuvers.
    - Chapter 08 - Airport Traffic Patterns.
    - Chapter 10 - Performance Maneuvers.
- Pilot's Handbook of Aeronautical Knowledge. (FAA-H-8083-25B)
    - Chapter 14 - Airport Operations.

Lesson Content (Ground)

Towered Airport Operations

- Ground operations:
    - The ATIS needs to be obtained as this will be used for an update on the airport/weather conditions and to let ground control know we have the current ATIS information.
    - Calling ground control:
        - Instructor acts as ATC.
        - Student: “Melbourne Ground, NxxxFT, at the south FIT run up area, with information X, ready to taxi, VFR to the south east”.
        - Give the student complex taxi instructions.
        - The student should have the airport diagram out, write the instructions, and read back the instructions.
        - Emphasize to the student to <u>always brief the taxi instructions!</u>
- Departure:

- Calling tower:
  - Instructor acts as ATC.
  - Student: "Melbourne Tower, NxxxFT, holding short of runway XX, ready for departure".
  - Give the student common Melbourne tower clearances:
    - "Back taxi approved, runway 23 cleared for takeoff, turn left on course".
    - "Runway 09L cleared for takeoff, on departure cross runway 5 midfield and join the right downwind".
    - "Runway 05 cleared for takeoff, maintain runway heading until midriver, then right turn to the south west".
  - Ensure that the student can properly explain what needs to be done in those scenarios.
  - Introduce line up and wait.
  - Review runway incursion avoidance.
- Departing Class D:
  - Frequency should not be changed until:
    - Clear of the airspace (laterally or vertically).
    - Frequency change approved.
- Returning for landing:
  - Descent/arrival checklist.
    - Make sure the student understands the importance of the approach brief.
  - Melbourne has three reporting points:
    - Radiation, Abandon, and Lake Washington.
  - Go over the entry procedures when reporting over the three reporting points.
    - Give different scenarios for each of the reporting point.
      - Ex: student is over abandon and requesting touch and goes (runway 09L, 09R, and 05 in use).
    - Give the student common Melbourne tower clearances:
      - "Enter a right base for runway 09L, remain 1-mile West of the interstate, report a 3 mile right base".
      - "Enter a left base for runway 27L, report the causeway".
      - "Cleared touch and go runway 09L, after departure, turn left crosswind prior to the VOR and remain in left closed traffic".
      - "Proceed to the numbers".
      - "Make short approach".
- After landing:
  - <u>Never</u> apply full breaks to make a taxiway, even if tower gave the instructions to exit on that said taxiway.
    - Tell tower "unable" and wait for new instructions. If none are given, exit on the next taxiway.

- Melbourne tower usually gives taxi instructions right after landing. If none are given, vacate the runway, taxi clear of the hold short line, and contact ground.
  - "Melbourne ground, NxxxFT is clear of runway 5 on Romeo, request taxi to the FIT ramp".
    - Emphasize to the student to <u>always brief the taxi instructions!</u>
- Define the terms "stay with me" and "monitor ground".

Radio Communications in Controlled Airspace

- Go over 91.123 with the student - compliance with ATC clearances and instructions.
- Class D:
  - Two-way radio communications before entering the airspace.
- Briefly introduce radio communications in class C and B:
  - Class C:
    - Two-way radio communications before entering the airspace.
    - Before taxiing and departing, pilots have to obtain a VFR clearance.
      - "Palm Beach clearance, NXXXFT requesting VFR clearance to Melbourne, altitude 3000 ft and type is a Piper Warrior ''.
      - "NXXXFT, Palm Beach clearance, cleared to Melbourne, after departure climb to 3000 ft and contact departure on 123.45".
  - Class B:
    - Clearance is required to enter the airspace.
    - Before taxiing and departing, pilots have to obtain a VFR clearance.
      - Similar to the example above.

Steep Turns, Engine Failure in Flight, Emergency Descent, Normal Traffic Pattern

- Quiz the student on the procedures and address any questions they might have.

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
  - Personal minimums.
  - FIT minimums.
  - Aircraft.
  - Airspace.
- Review the students planning with a focus on:
  - Weather, weight and balance, performance, flight planning, and fuel calculations.
  - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
  - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
  - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

<u>**Lesson Content (Dual)**</u>

Preflight Inspection
- Ensure that the student conducted a thorough preflight inspection.

Visual Scanning and Collision Avoidance
- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Normal/Crosswind Takeoff
- The student should conduct the takeoff.
- Ensure the runway entry check and the lineup check are completed.
- Ensure proper crosswind correction is maintained.

Review Previous Exercises as Required
- Review any areas the student might need help with by utilizing the pertinent lesson plans.

Steep Turns
- <u>Student says, student does:</u>
    - Verify correct procedure.
- Points of Emphasis.
    - Area clearance.
    - Coordination.
    - Trim and power usage.

Ground Reference Maneuvers
- Turns around a point:
    - <u>Student says, student does.</u>
        - Verify correct procedure.
    - Points of Emphasis.
        - Entry procedures: downwind, sufficient space to set up, and proper spacing from point.
        - Bank variation.
        - Choosing an adequate and safe visual reference point.
        - Provide feedback as needed throughout the maneuver.
- S-turns:
    - <u>Student says, student does.</u>
        - Verify correct procedure.
    - Points of Emphasis.
        - Entry procedures: downwind, sufficient space to set up, and proper spacing from point.
        - Bank variation.
        - Choosing an adequate and safe visual reference point.

- Provide feedback as needed throughout the maneuver.

Engine Failure in Flight

- Student says, student does.
  - Verify correct procedure.
- Points of Emphasis.
  - Trimming for V_{glide}.
  - Wire strike avoidance.
  - Choice of landing point.

Emergency Descent

- Student says, student does.
  - Verify correct procedure.
- Points of Emphasis.
  - Trim.
  - Area clearance.
  - Ensure the student does not exceed V_{FE}.

Normal Traffic Pattern and Powered Approach at Towered Airport

- After all line items have been satisfied, proceed back to KMLB.
- Ensure that the descent/arrival checklist is completed.
- Before telling the student whether to go to radiation or abandon, see if they can figure out which reporting point is best depending on the runways in use.
- The student will be in charge of all of the radio calls with tower and ground.
  - Emphasize the use of saying "say again" if the instructions are not clear.
  - Emphasize the use of saying "unable" if the instructions are not safe or cannot be followed.
  - Help if needed.
- Traffic patterns are to be completed by the student with minimal instructor guidance.
- Provide guidance on using visual references (Sarno road, VOR, etc.).

Normal/Crosswind Landing

- The student should conduct the landing.
- By this lesson, the student should be able to conduct a safe landing with minimal instructor guidance.

Go-Around/Rejected Landing

- Have the student conduct a go-around during one of the traffic patterns.
- Ensure that the correct procedure is conducted.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 117.
- Assign homework:
    - Review emergency procedures and checklists.
    - Review ground reference maneuvers.
    - Review towered operations → listen to liveatc.net.

Completion Standards

This lesson is complete when the student is able to:

1. Correctly perform previously learned maneuvers with little instructor guidance.
2. Demonstrate proper sequential checklist procedures for all normal procedures without instructor guidance.
3. Demonstrate proper sequential checklist procedures for emergency procedures with instructor guidance.
4. During ground reference maneuvers, fly a predetermined ground track using specific ground references, understand the effects of wind, correct for wind drift and maintain altitude ±150 ft, and airspeed ±10 knots, maximum bank 45° with little instructor guidance.
5. Conduct emergency procedures with little instructor guidance.
6. Maintain the recommended traffic pattern altitude ±150 ft, recommended airspeed +5/-0 knots and correct for wind drift without instructor guidance.